The research program of the Center for Economic Studies (CES) produces a wide range of theoretical and empirical economic analyses that serve to improve the statistical programs of the U.S. Bureau of the Census. Many of these analyses take the form of CES research papers. The papers are intended to make the results of CES research available to economists and other interested parties in order to encourage discussion and obtain suggestions for revision before publication. The papers are unofficial and have not undergone the review accorded official Census Bureau publications. The opinions and conclusions expressed in the papers are those of the authors and do not necessarily represent those of the U.S. Bureau of the Census. Republication in whole or part must be cleared with the authors.

CONTRIBUTIONS TO HEALTH INSURANCE PREMIUMS: WHEN DOES THE EMPLOYER PAY 100 PERCENT?

by

Alice Zawacki *
U.S. Bureau of the Census

and

Amy K. Taylor *
Agency for Healthcare Research and Quality

CES 05-27 **December, 2005**

All papers are screened to ensure that they do not disclose confidential information. Persons who wish to obtain a copy of the paper, submit comments about the paper, or obtain general information about the series should contact Sang V. Nguyen, Editor, Discussion Papers, Center for Economic Studies, Washington Plaza II, Room 206, Bureau of the Census, Washington, DC 20233-6300, (301-763-1882) or INTERNET address snguyen@ces.census.gov.

Abstract

We identify the characteristics of establishments that paid 100 percent of health insurance premiums and the policies they offered from 1997-2001, despite increased premium costs. Analyzing data from the MEPS-IC, we see little change in the percent of establishments that paid the full cost of premiums for employees. Most of these establishments were young, small, single-units, with a relatively high paid workforce. Plans that were fully paid generally required referrals to see specialists, did not cover pre-existing conditions or outpatient prescriptions, and had the highest out-of-pocket expense limits. These plans also were more likely than plans not fully paid by employers to have had a fee-for-service or exclusive provider arrangement, had the highest premiums, and were less likely to be self-insured.

KEYWORDS: employer-sponsored health insurance, contributions, premiums

* The analysis and results presented in this paper are attributable to the authors and do not necessarily reflect concurrence by the Center for Economic Studies, U.S. Census Bureau or the Agency for Healthcare Research and Quality. This paper has undergone a more limited review by the Census Bureau than official Census publications. It has been screened to ensure that no confidential data are revealed.

INTRODUCTION

While the health care sector experienced low inflation during the mid-1990s, in 2001 health care spending per capita rose 10 percent, which represented the largest increase in a decade.[1] These rising medical care costs are associated with an increase in health insurance premiums.[2] Higher insurance costs create budgetary concerns for employers, since health insurance benefits represent such a large component of employee compensation.

Employers are the major source of health insurance in this country, and they often try to pass some of the increased insurance costs on to employees through higher copayments, coinsurance, and deductibles[3] and/or higher employee contributions for health insurance premiums.[4] Employer contribution levels vary across firms, and tight labor markets, unionization, and political pressures have often influenced the share of premiums paid by employers and employees.[5] The exclusion of employer paid health insurance from employee taxable income is another factor that influences employer

---

[1] Strunk, B.C., P.B. Ginsburg, J.R. Gabel, "Tracking Health Care Costs: Growth Accelerates Again in 2001," *Health Affairs* Web Exclusives (2002): W299.

[2] Gabel, J., L. Levitt, J. Pickreign, H. Whitmore, E. Holve, D. Rowland, K. Dhont, S. Hawkins, "Job-Based Health Insurance in 2001: Inflation Hits Double Digits, Managed Care Retreats," *Health Affairs* 20 (2001): 180-186.

[3] Robinson, J.C., "Renewed Emphasis on Consumer Cost Sharing In Health Insurance Benefit Design," *Health Affairs* Web Exclusive (2002): W139-W154. Trude, S., J.B. Christianson, C.S. Lesser, C. Watts, A.M. Benoit, "Employer-Sponsored Health Insurance: Pressing Problems, Incremental Changes," *Health Affairs* 21 (2002): 66-75. Marquis M.S. and S.H. Long, "Prevalence of Selected Employer Health Insurance Purchasing Strategies in 1997," *Health Affairs* 20 (2001a): 220-230.

[4] Schiff, M., M. Schuster, S. Bachman, A. Lischko, "Employee Input and Health Care Cost-Containment Strategies," *Managed Care Interface* (2003): 20-24. Gabel, J., L. Levitt, E. Holve, J. Pickreign, H. Whitmore, K. Dhont, S. Hawkins, D. Rowland, "Job-Based Health Benefits in 2002: Some Important Trends," *Health Affairs* 21: (2002): 143-151. Gabel, J., G. Claxton, E. Holve, J. Pickreign, H. Whitmore, K. Dhont, S. Hawkins, D. Rowland, "Health Benefits in 2003: Premiums Reach Thirteen-Year High as Employers Adopt New Forms of Cost Sharing," *Health Affairs* 22: (2003): 117.

[5] Buchmueller, J.C., J. DiNardo, R.G. Valletta, "Union Effects on Health Insurance Provision and Coverage in the United States," *Industrial and Labor Relations Review* 55 (2002): 610-627. Montagne, C, "Bargaining Health Benefits in the Workplace: An Inside View," *The Milbank Quarterly* 80 (2002): 547-567. Marquis M.S. and S.H. Long, "Employer Health Insurance and Local Labor Market Conditions," *International Journal of Health Care Finance and Economics* 1 (2001b): 273-292. Rossiter, L.F. and A.K. Taylor, "Union Effects on the Provision of Health Insurance," *Industrial Relations* 21 (1982): 167-177.

contributions to health insurance premiums.[6] This paper will examine those establishments that continued to pay 100 percent of health insurance premiums for their employees, despite increased medical care and insurance costs. Research has shown that workers are more sensitive to out-of-pocket than total health insurance premiums,[7] so that 100 percent payment by employers would be particularly valued. While employee contributions are rising[8] and although economists debate the share of employer paid premiums that are ultimately shifted back to workers in the form of reduced wages, the goal of this analysis is to identify the characteristics of those establishments that paid 100 percent of health insurance premiums and the policies they offered.

We analyze data from 1997 through 2001 from the Medical Expenditure Panel Survey – Insurance Component (MEPS-IC), which collects establishment level data on the health insurance plans offered to employees, as well as establishment and workforce characteristics. We focus on establishments that offered health insurance and contributed 100 percent of the premium cost for at least one of the plans they offered. This paper begins by looking at the percent of establishments in the United States that offered health insurance at no cost to their employees for the period 1997 through 2001. Next, we describe the characteristics of those establishments that contributed 100 percent of the premium cost for employees. Finally, we examine the characteristics of plans with

[6] Taylor, A.K. and G.R. Wilensky, "The Effect of Tax Policy on Expenditures for Private Health Insurance," *Market Reform and Health Care: Current Issues, New Directions and Strategic Decisions,* Jack Myer, Editor. American Enterprise Institute, 1983.
[7] Cooper, P. and J. Vistnes, "Workers' Decisions to Take-Up Offered Health Insurance Coverage: Assessing the Importance of Out-of-Pocket Premium Costs," *Medical Care* Vol. 41, No. 7 (July 2003): p. 35-43.
[8] Crimmel, B.L. "Employee Contributions to Employer-Sponsored Health Insurance Coverage, 1997 v. 2002." Statistical Brief No. 55, September 2004, Agency for Healthcare Research and Quality, Rockville, MD. http://www.meps.ahrq.gov/papers/st55/stat55.pdf

premiums entirely paid by employers and compare these to the characteristics of plans not fully paid by the employers.

DATA AND METHODS

The Medical Expenditure Panel Survey (MEPS) Insurance Component (IC) is a survey of employers that is done annually, and covers state and local governmental units as well private-sector establishments. It is sponsored by the Agency for Healthcare Research and Quality and is conducted by the U.S. Census Bureau. First conducted in 1996, the MEPS – IC provides estimates of employer-sponsored insurance at both the national and the State level for 40 states in any given year.

In this survey establishment refers to a particular workplace or location, while a firm is a business entity consisting of one or more business establishments under common ownership or control.[9] Because the unit of analysis in the MEPS – IC is the establishment, the analysis in this paper is establishment-based. In addition, laws that regulate health insurance vary by state and this can result in establishments within the same firm offering different benefits. In fact we find that establishments within the same firm do not always make the same level of contributions. The MEPS – IC provides data on employer characteristics, including measures for establishment size, establishment ownership type, industry, location, and the age of the firm. Information on the percent of the workforce that is female, over the age of 50, unionized, and earning low/medium/high wages is also included.

[9] Branscome, J.M. and B.L. Crimmel, "Changes in Job-Related Health Insurance, 1996-99," Rockville (MD): Agency for Healthcare Research and Quality (2002), MEPS Chartbook No. 10, AHRQ Publication No. 02-0030. Cohen, J.W., et al., "The Medical Expenditure Panel Survey: A National Health Information Resource," *Inquiry* 33 (Winter 1996/97): 373-389.

Each establishment or reporting unit in the MEPS – IC provides information about the health insurance plans they offer. These plans represent insurance contracts that provide coverage for health care services to an employee and/or retiree for an agreed-upon fee (premium) for a defined benefit period. Some plans have self-insured indemnification, which means that the employer is assuming the risk for the employees' medical expenses. This is in contrast to purchased plans in which the health insurance company assumes the financial risk for the enrollee's medical claims.[10] Some plans may offer single coverage that only covers the employee. When the employee and the employee's family are covered, the plan is said to offer family coverage.[11] The MEPS – IC collects data on premiums for single and family coverage, contributions by employers and employees, provider type, plan enrollment, deductibles, and copayments.

For this analysis, we used the MEPS – IC data on private-sector establishments from 1997 to 2001. The sample size was between 25,000 and 30,000 units for each of the five years. The estimates were weighted to be nationally representative, and tests of statistical significance accounted for the survey design of the MEPS – IC. All differences discussed in this article were statistically significant at the .05 level.

RESULTS

Employer – Sponsored Health Insurance: Establishments. We begin by looking at the percent of establishments that paid the full premium cost for either single coverage and/or family coverage. We separated establishments into those that offered only one health insurance plan and those that offered more than one plan. As Figure 1 shows, both

[10] Academy Health, *Glossary of Terms Commonly Used in Health Care*, 2003 Edition.
[11] Branscome, J.M. and B.L. Crimmel, 2002.

types of establishments were significantly more likely to pay 100 percent for single coverage than for family coverage. We also see that the percent of establishments paying the full cost changed very little from 1997 to 2001.

It is important to keep in mind here that the distribution of establishments is not the same as the distribution of workers. Because many establishments are single-units and small in terms of the number of persons employed, the health benefit decisions made by these employers impact a small percentage of the workforce. For example, in 1997 almost 60 percent of establishments that offered one plan paid 100 percent of the premium cost for single coverage, but only 40 percent of employees at these establishments had fully paid single coverage plans (data not shown).[12] Similar to findings for establishments, a significantly higher percentage of workers in all years (1997 through 2001) had premiums fully paid by employers for single coverage than for family coverage (data not shown).[13]

Next, we describe the characteristics of the establishments that paid 100 percent of premium costs. First, organizational characteristics of employers, including size, non-profit status, industry, location of the establishment and age of the firm, were examined. Second, we focused on characteristics of the establishment's workforce, including the percent of workers that were low wage earners, part-time workers, and unionized.

Figure 2 shows that while larger establishments were more likely to offer insurance, smaller establishments were more likely to pay the full premium cost for single and for family coverage if they offered health insurance at all. We found that single-unit establishments were also more likely to pay 100 percent than multi-unit

[12] Zawacki, A. and A.K. Taylor, " Contributions to Health Insurance Premiums: When Does the Employer Pay 100 Percent?" Presented at Academy Health's 2004 Annual Research Meeting, San Diego, June 2004.
[13] Ibid.

establishments (data not shown).[14] The percent of establishments that paid 100 percent by establishment characteristics for 1997 through 2001 is shown in Table 1. Establishments that are part of firms operating for less than 5 years were significantly more likely to pay 100 percent for at least one plan than establishments in firms operating for more than 20 years. We also found that nonprofit establishments were more likely than for-profit establishments to pay 100 percent of the premium cost for at least one plan.

It is also interesting to look at the characteristics of the workforce in establishments paying the full premium cost of health insurance for their employees (Table 1). Establishments with 50 percent or more of their workforce earning a low wage were significantly less likely to pay 100 percent for at least one plan than establishments with a higher paid workforce. Low wages were defined in the MEPS – IC as less than $6.50 per hour in 1997 through 1999 and less than $9.50 per hour in 2000 and 2001. Establishments with a higher percentage of part-time workers were less likely to pay 100 percent for family coverage. Finally, in establishments that offered more than one plan, those that were more unionized were more likely to pay 100 percent of the premiums for both single and family coverage than establishments with less than 25 percent of their employees belonging to a union.

Establishments in the agriculture/fishing/forestry and construction industries were less likely to offer insurance than those in other industries (data not shown).[15] Establishments in these industries, however, were more likely to pay 100 percent when they did offer insurance. This was true for both single and family coverage.

[14] Ibid.
[15] Ibid.

Establishments in the retail industry were least likely to pay 100 percent for single or family coverage. We found that establishments in the retail and financial industries that offered more than one plan were less likely to pay 100 percent for health insurance than establishments in other industries.

The percent of establishments that paid 100 percent of health insurance premiums also varied by location (data not shown).[16] Establishments in the south were less likely than those in than other regions to have paid 100 percent for single or family coverage, whether the establishment offered one plan or more than one plan. Establishments in the west that offered one plan with single coverage were significantly more likely to pay 100 percent than establishments in other regions. Establishments located in the northeast, which offered only one plan, were significantly more likely to pay 100 percent for family coverage than those in the rest of the U.S.

Employer – Sponsored Health Insurance: Plans

The previous section looked at establishments that offered health insurance and paid 100 percent of the premiums for at least some of the plans that they offered. We now look at the characteristics of the plans that were fully paid by employers. These characteristics included the plan's provider type arrangement, whether referrals were required, self-insured indemnification, coverage for pre-existing conditions, coverage for outpatient prescriptions, premiums, and out-of-pocket expenses.

This analysis considered three different types of plans characterized by provider type arrangements[17] as identified in the MEPS – IC survey. The first category of plans

[16] Ibid.
[17] Academy Health, 2003.

included health maintenance organizations (HMO), independent physicians associations (IPA), and exclusive provider organizations (EPO). This type of plan required that enrollees go to providers associated with the plan for all non-emergency care in order for the costs to be covered. The second category of provider arrangements included most fee-for-service plans. If enrollees may go to providers of their choice with no cost incentives to use a particular group of providers, this was considered a plan with any provider. Preferred provider organizations (PPO) and point-of-service plans fell into the third category, which was a mixture of preferred and any provider.

Figure 3 shows provider type by employer contributions for single coverage in 2001. The findings are very similar for family coverage. The majority of plans offered at establishments providing health insurance had a mixed provider type arrangement. However, fully paid plans were more likely to have exclusive provider or any provider arrangements than plans which were not paid 100 percent.

Table 2 shows health plan characteristics by employer contributions for 2001. First, we looked at premiums for single and family coverage, which were categorized as low, medium, or high. Premiums in the low category were greater than $0 and less than or equal to $2000 for single coverage and greater than $0 and less than or equal to $5000 for family coverage. Premiums in the middle category fell between $2000 and $5000 for single coverage and $5000 and $10,000 for family coverage. Premiums in the high category were greater than $5000 for single coverage and greater than $10,000 for family coverage. We found that all plans that were paid 100 percent were more likely to have premiums in the highest group.

Table 2 also shows that plans that were fully paid by employers were less likely to be self-insured. Next, we looked at whether plans required a referral to see a specialist. Plans that were paid 100 percent by the employer were more likely to require referrals to see a specialist than plans that were not fully paid.

We then looked at whether plans covered pre-existing conditions or outpatient prescriptions. Some health insurance plans restrict coverage for medical or health conditions which exist prior to enrollment in the plan.[18] Plans that were paid 100 percent were less likely to cover pre-existing conditions than plans requiring employee contributions towards premiums, except for fully paid family plans offered by establishments offering more than one plan. Plans with either single or family coverage that were paid 100 percent at establishments offering only one plan were less likely to cover outpatient prescriptions than other plans.

Finally, we looked at maximum annual out-of-pocket expenses for the health insurance plans. Annual limits on these expenses were categorized as low, medium, or high. Maximum expense limits that were greater than zero and less than $1000 were categorized as low. If these expense limits were more than $1000 and less than or equal to $5000, they were grouped in the medium category. The high category included maximums that were greater than $5000. Fully paid plans were more likely to be in the highest category than other plans.

DISCUSSION/SUMMARY

Today's headlines reflect concerns over rising health care costs, employers' struggles with containing the growing cost of health insurance premiums, and the number

[18] Ibid.

of working Americans without health insurance. Despite these concerns many employers continued to pay the full cost of health insurance for their employees from 1997 to 2001.

We found that the provision of employer-sponsored health insurance at no cost to employees varied according to specific organizational, workforce, and plan characteristics. Most of these establishments that paid 100 percent of premium costs for their employees were young, small, single-units, with a higher paid workforce. Across plans at all establishments offering one or more plan, those that were fully paid were more likely than plans not fully paid by employers to have a fee-for-service or any provider arrangement, had the highest premiums, and were less likely to be self-insured. Plans that were fully paid by employers tended to be less generous in terms of benefits. Plans that were paid 100 percent by employers generally required referrals to see specialists, did not cover pre-existing conditions or outpatient prescriptions, and had the highest out-of-pocket expense limits.

Finding that small establishments were more likely to pay 100 percent of premium costs for at least one health plan and that fully paid plans were more likely to have the highest premium costs may relate to insurers' and employers' worries regarding adverse selection. Insurers charge higher premiums for plans with a low percentage of the workforce enrolled, because they are concerned that only the sickest employees with the highest health care costs are participating in the plan.[19] Small establishments, therefore, may encourage enrollment by fully contributing towards the cost of the plan in order to lower premium costs. Paying the full cost of an employee's insurance premiums

[19] Gabel, J.R., J.D. Pickreign, "Risky Business: When Mom and Pop Buy Health Insurance for Their Employees," New York (NY): The Commonwealth Fund (2004), Issue Brief #772. Cave, D.G. and L.J. Tucker, "How Will Employers Manage Employee Risk Selection Among Health Plans in the 1990s?" *Benefits Quarterly* 6 (1990): 1-13.

provides an incentive to participate in the firm's health insurance plan. Researchers have shown that employee cost sharing affects enrollment in health plans.[20] Further, small establishments might not have a choice because some insurers require that all workers be covered as a condition of offering insurance at all.[21] In addition, state laws relating to the sale of small employer health insurance policies sometimes allow insurers to establish employer contribution rules in order to meet minimum participation requirements.[22]

We have seen that the percent of establishments that paid the full cost of health insurance premiums for employees changed very little between 1997 and 2001. As health care costs inevitably rise in the future, it will be interesting to see if this behavior continues. In future research, we also will be pursuing the issue of what is happening to the characteristics of fully paid health insurance plans and the impact of state laws regulating the market for small employer health insurance. We will continue to monitor the benefits to employees and see whether health insurance is becoming more costly in terms of higher copayments, coinsurance, and deductibles when employers pay 100 percent of the premium cost.

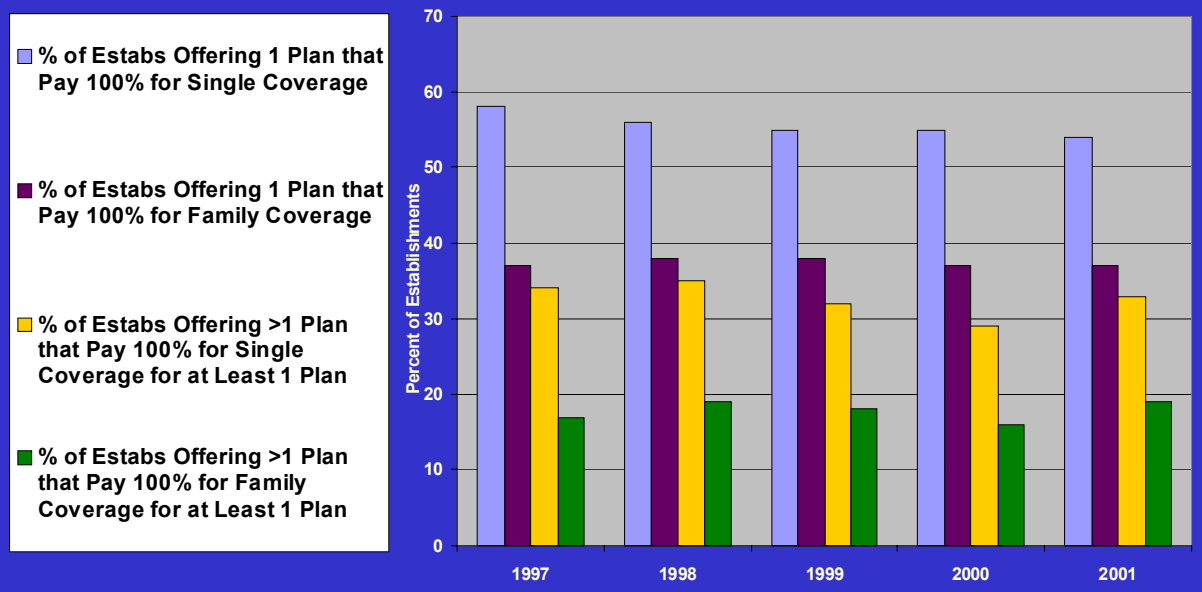
---

[20] Cutler, D.M., "Employee Costs and the Decline in Health Insurance Coverage," Cambridge (MA): National Bureau of Economic Research (2002), NBER Working Paper Series 9036. Gabel, J.R., J.D. Pickreign, H.H. Whitmore, C. Schoen, "Embraceable You: How Employers Influence Health Plan Enrollment," *Health Affairs* 20 (2001): 196-208. Rice, T., K. Desmond, N. Purat, "Dark Clouds in Pleasantville: Trends in Job-Based Health Insurance, 1996-1998," Presentation at the Association of Health Services Research annual meeting, Chicago, June 1999. Cooper, P.F. and B.S. Schone, "More Offers, Fewer Takers for Employment-Based Health Insurance: 1987 and 1996," *Health Affairs* 16 (1997): 142-149.

[21] Blumberg, L.J., "Who Pays for Employer-Sponsored Health Insurance?" *Health Affairs* 18, no. 6 (1999): 58-61.

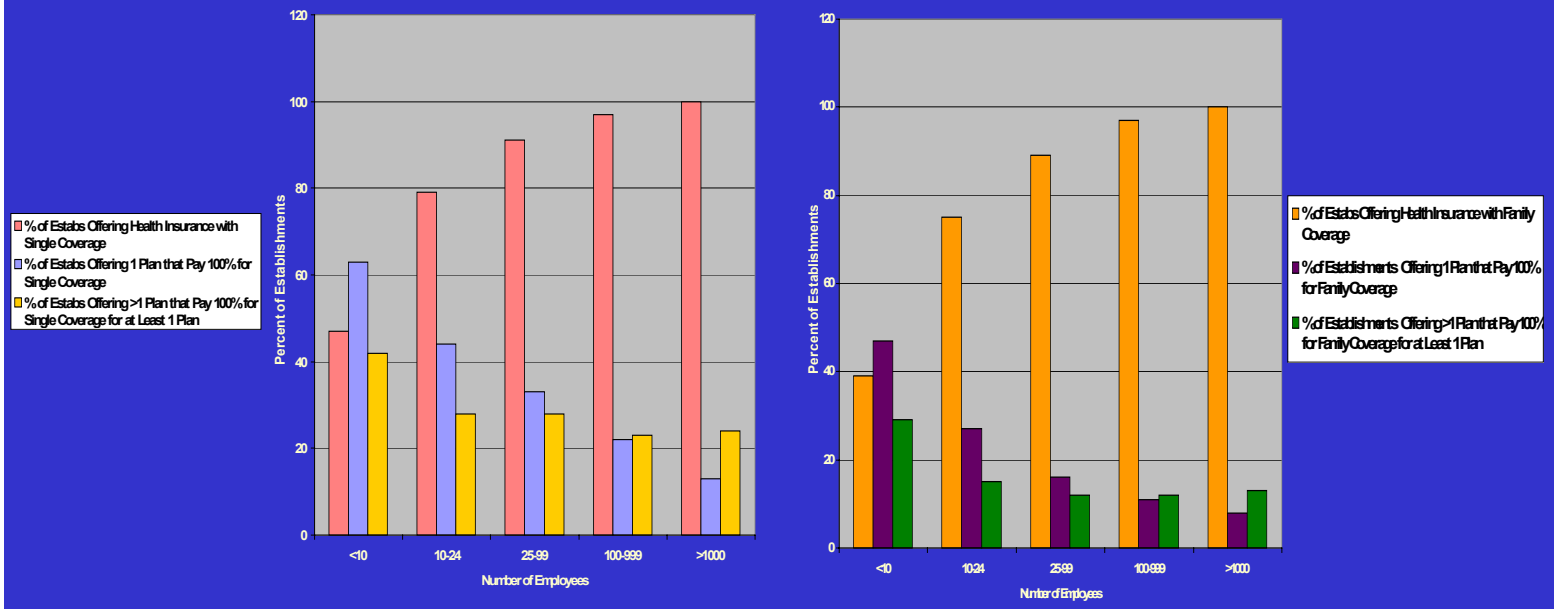
[22] http://www.dmhc.ca.gov/library/statutes/knox-keene/html/__1357_03_marketing_of_plans_to_small_employers_participation.htm; http://oci.wi.gov/sm_emp/h_reqsmemp.htm

Figure 1. Percent of Establishments that Pay 100% of Premium for at Least One Plan



Source: Medical Expenditure Panel Survey-Insurance Component (List Sample), sponsored by the Agency for Healthcare Research and Quality.

Figure 2. Percent of Establishments that Pay 100% by Size (2001)



Source: Medical Expenditure Panel Survey-Insurance Component (List Sample), sponsored by the Agency for Healthcare Research and Quality.

Table 1. Percent of Establishments that Pay 100 Percent by Establishment Characteristics (2001)

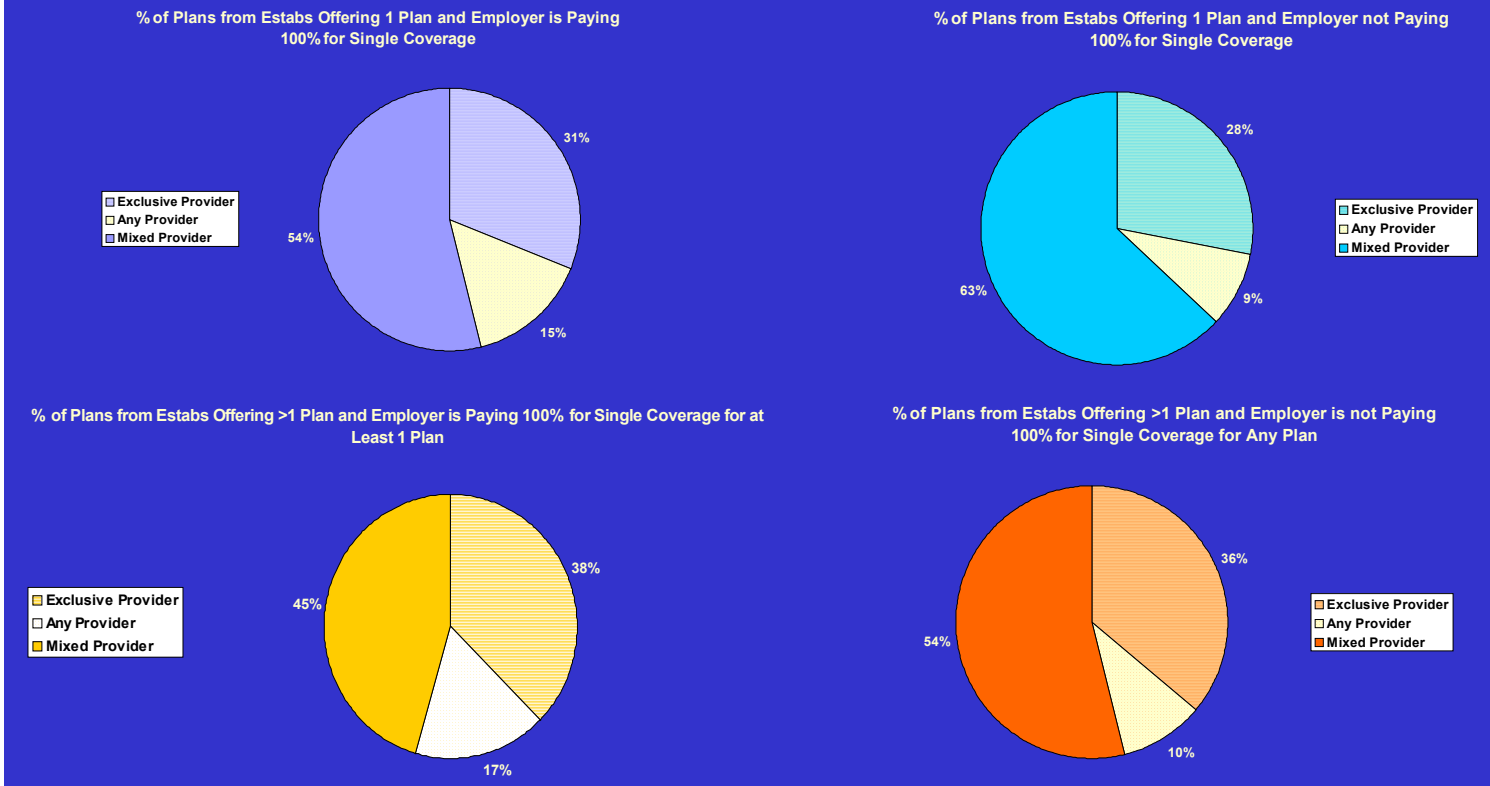
	% of Estabs Offering 1 Plan that Pay 100% for Single Coverage	% of Estabs Offering 1 Plan that Pay 100% for Family Coverage	% of Estabs Offering >1 Plan that Pay 100% for Single Coverage for at Least 1 Plan	% of Estabs Offering >1 Plan that Pay 100% for Family Coverage for at Least 1 Plan
Age of firm				
< 5 years	61	45	56	36
5-9 years	58	44	55	33
10-20 years	59	42	55	33
> 20 years	53	35	36	22
Non-profit status				
Non-profit	63	42	49	28
For profit	53	36	32	19
Low wage earners				
≥ 50% of workforce	46	33	26	15
< 50% of workforce	61	41	47	28
Part-time workers				
≥ 50% of workforce	49	37	21	14
< 50% of workforce	55	36	36	21
Union				
≥ 25% of workforce	65	54	60	48
<25% of workforce	54	36	32	17

Source: Medical Expenditure Panel Survey-Insurance Component (List Sample), sponsored by the Agency for Healthcare Research and Quality.

Figure 3. Provider Type by Employer Contributions for Single Coverage (2001)

100%

Not 100%



Source: Medical Expenditure Panel Survey-Insurance Component (List Sample), sponsored by the Agency for Healthcare Research and Quality.

Table 2. Health Plan Characteristics by Employer Contributions (2001)								
	One Plan				More Than One Plan			
	Single Coverage		Family Coverage		Single Coverage		Family Coverage	
	Employer Paying 100%	Employer Not Paying 100%	Employer Paying 100%	Employer Not Paying 100%	Employer Paying 100%	Employer Not Paying 100%	Employer Paying 100%	Employer Not Paying 100%
	Percent of Plans							
Premiums								
Low	18	18	17*	14	16	15	12	9
Medium	72*	76	63*	70	75*	82	72*	78
High	10*	6	20*	16	9*	3	16*	12
Self-insured	13*	29	13*	27	33*	53	38*	50
Referral not required	53*	57	53*	58	46*	56	45*	55
Cover pre-existing conditions	54*	60	54*	60	69*	74	71	74
Cover outpatient Rx	93*	96	93*	96	95	96	95	96
Maximum out-of-pocket expenses								
Low	25	27	10*	7	21	18	5	5
Medium	38*	41	39*	49	39*	44	36*	45
High	37*	32	51*	44	40	38	59*	51

* Difference between employer paying 100 percent and not paying 100 percent is significant at the .05 level

Source: Medical Expenditure Panel Survey-Insurance Component (List Sample), sponsored by the Agency for Healthcare Research and Quality.